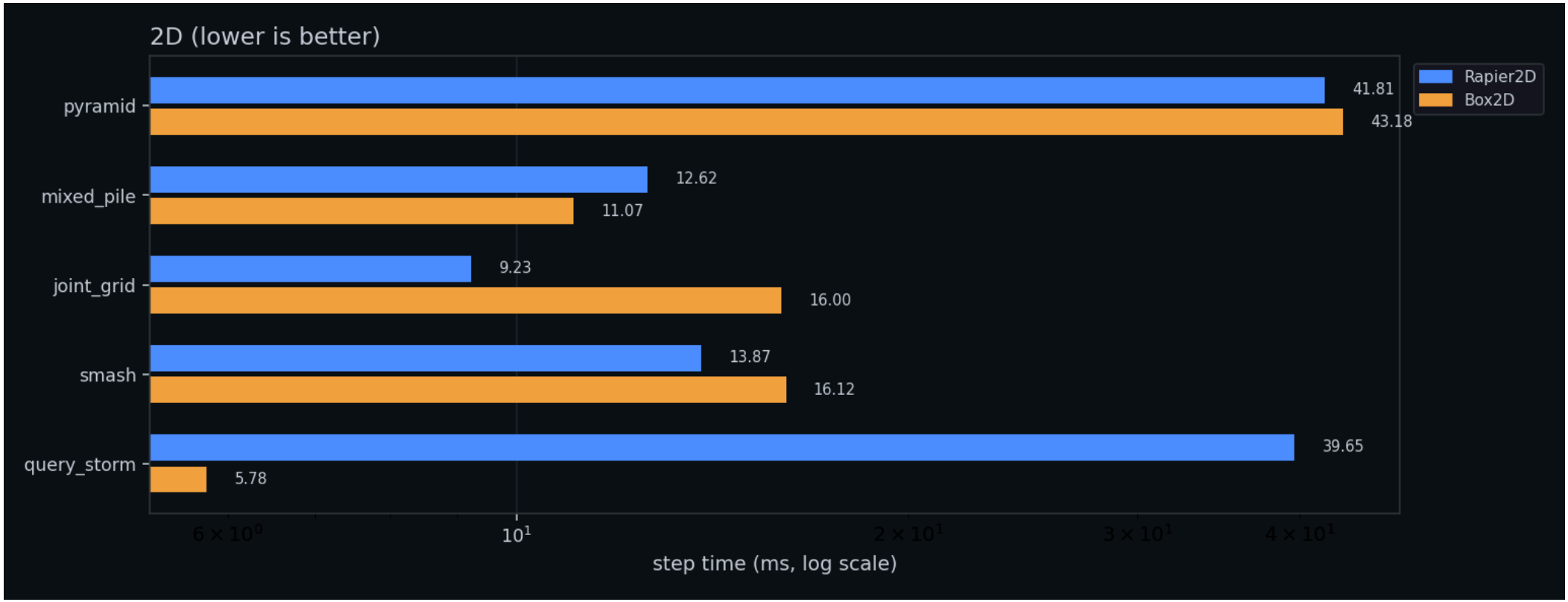
Godot physics engine benchmark

Godot 4.7-stable (official) · 150 timed steps · 60 Hz · single run · lower is better.

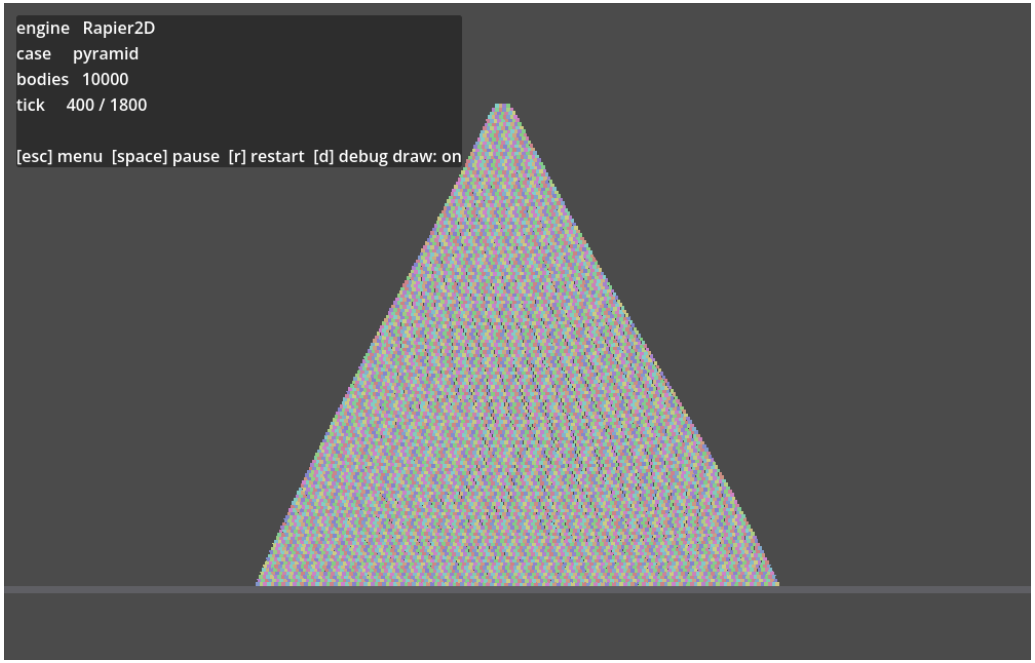
2D (lower is better)



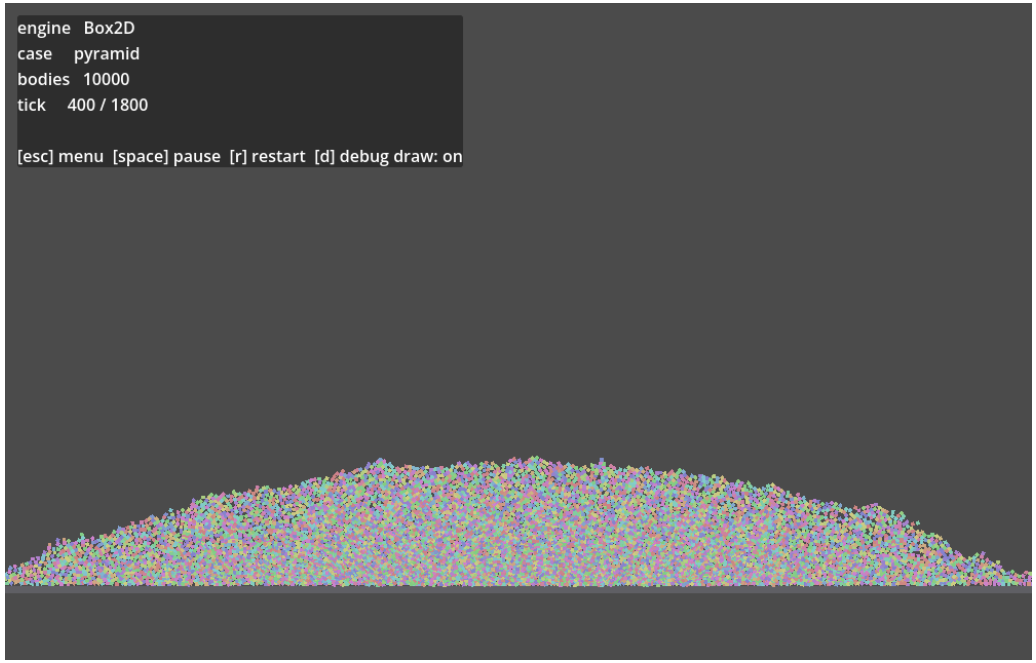

2D — relative to Rapier2D
pyramid
mixed_pile
joint_grid
smash
query_storm
Box2D
10^0
step time relative to Rapier2D (1.0 = same; >1 slower than Rapier2D)

`pyramid` (2d)

Rapier2D



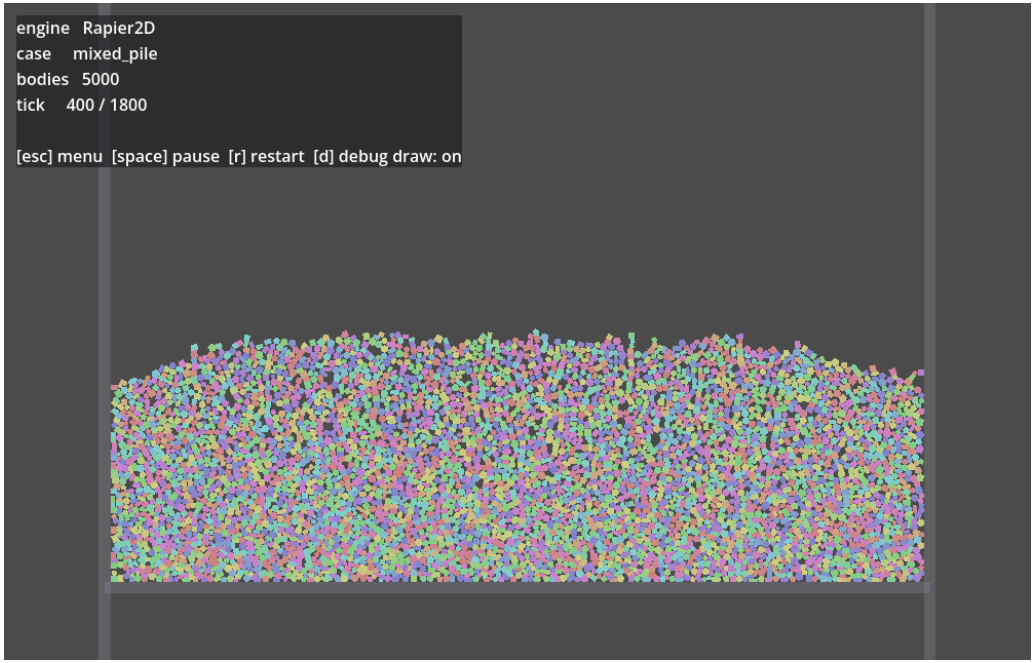
Box2D



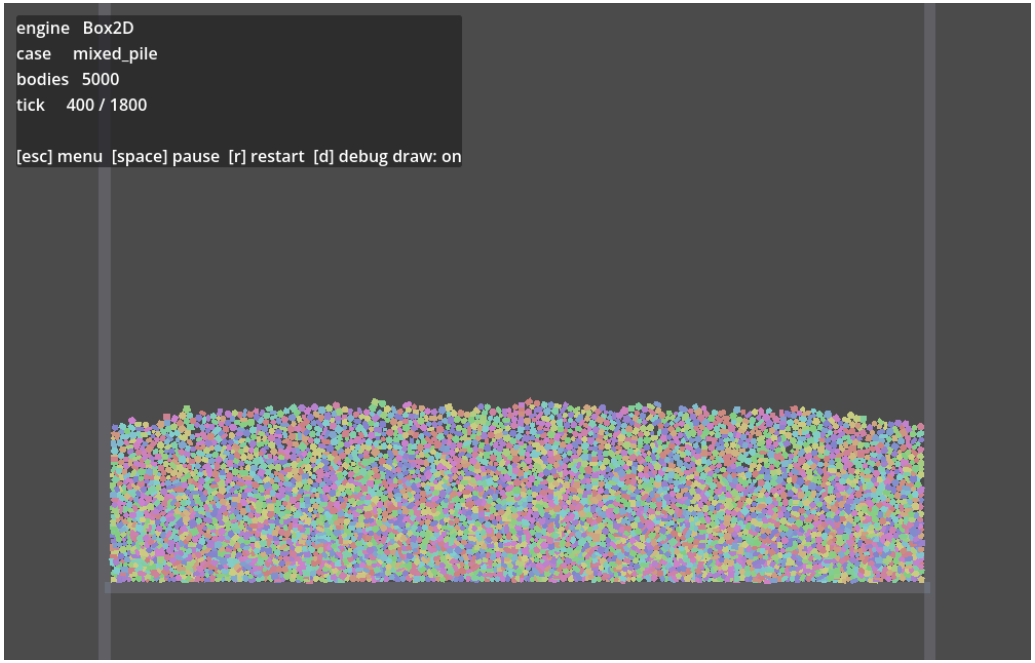
engine	step p50	step p99	vs Rapier2D
Rapier2D	**41.81 ms**	94.50 ms	—
Box2D	43.18 ms	100.56 ms	1.03x

`mixed_pile` (2d)

Rapier2D



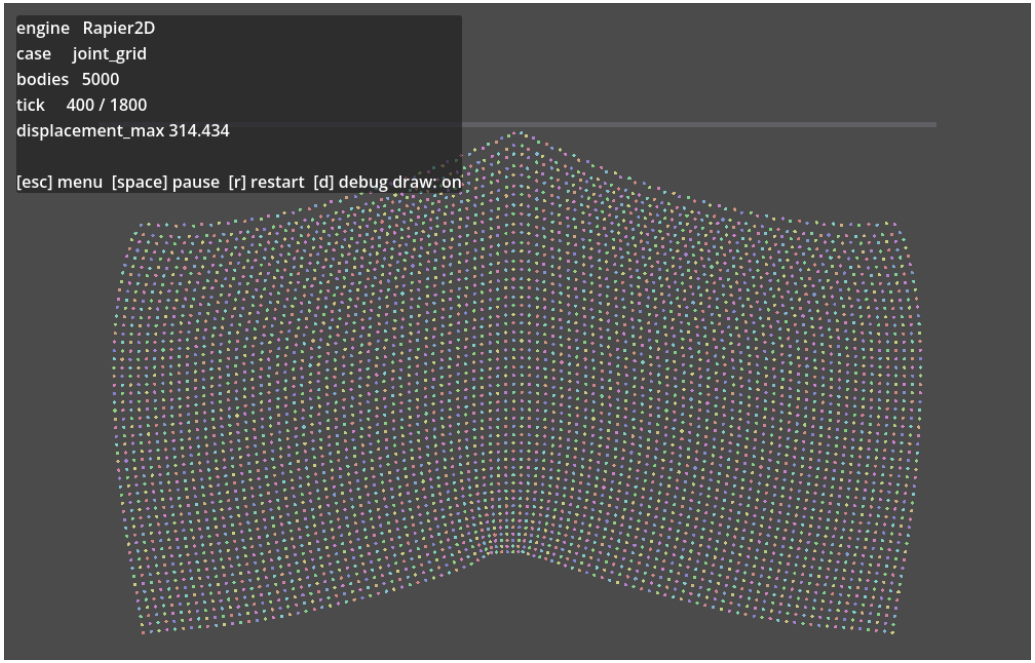
Box2D



engine	step p50	step p99	vs Rapier2D
Box2D	**11.07 ms**	20.66 ms	0.88x
Rapier2D	12.62 ms	26.89 ms	—

`joint_grid` (2d)

Rapier2D



Box2D

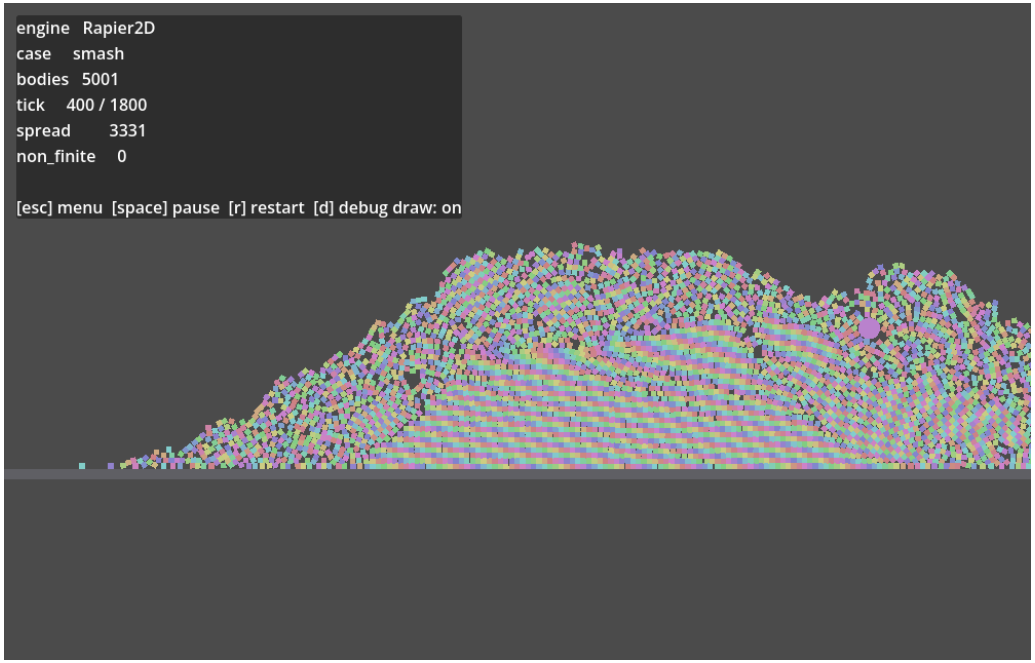


engine	step p50	step p99	vs Rapier2D
Rapier2D	**9.23 ms**	11.10 ms	—
Box2D	16.00 ms	34.30 ms	1.73x

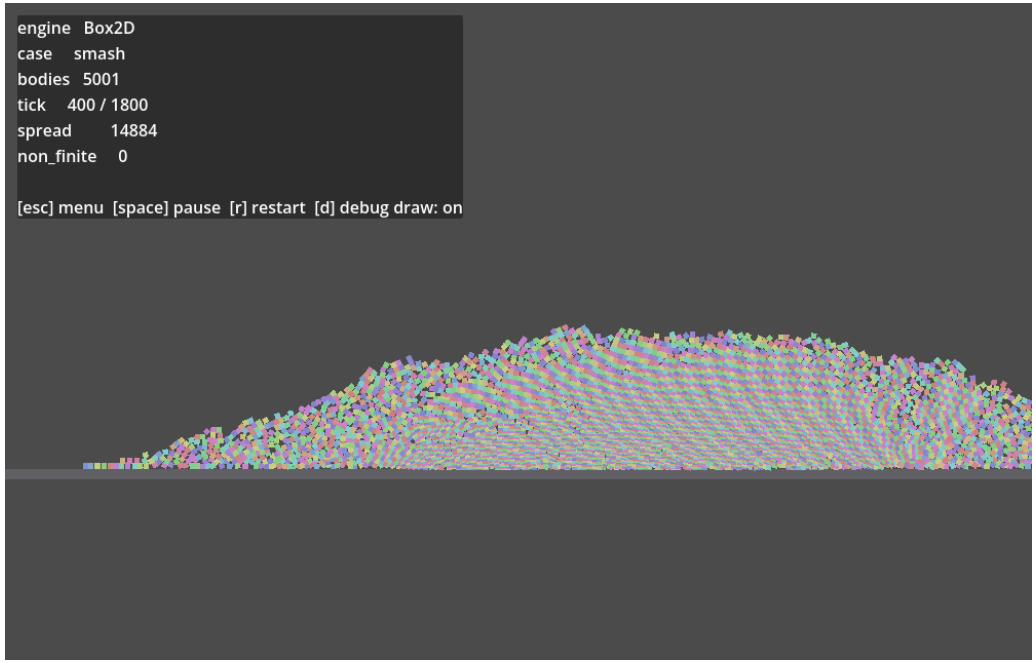
engine	`displacement_max`
Box2D	inf
Rapier2D	119.8

`smash` (2d)

Rapier2D



Box2D

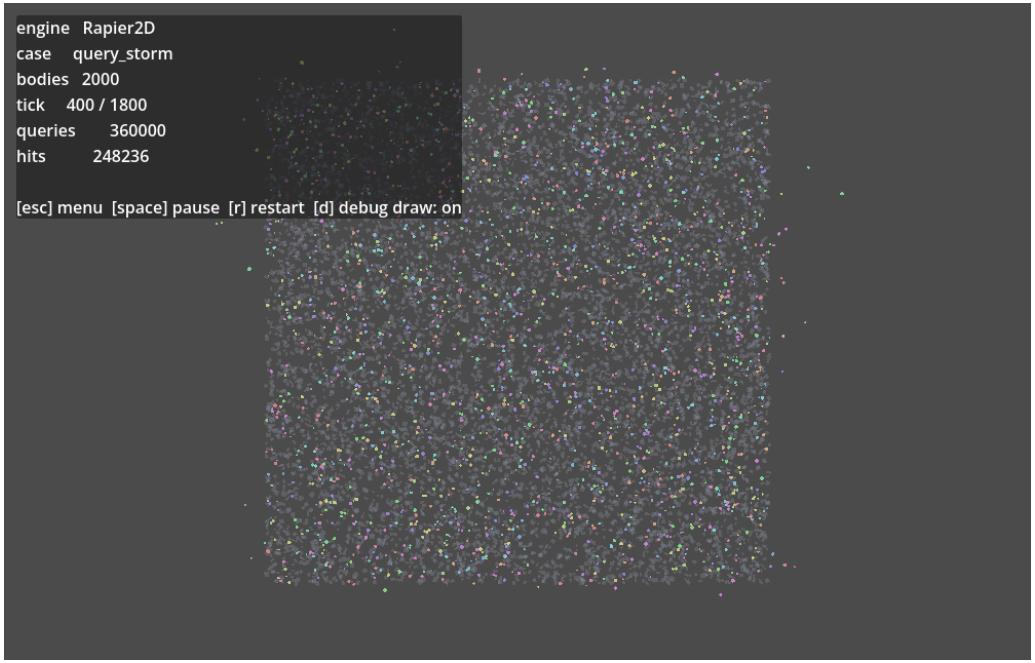


engine	step p50	step p99	vs Rapier2D
Rapier2D	**13.87 ms**	22.02 ms	—
Box2D	16.12 ms	25.22 ms	1.16x

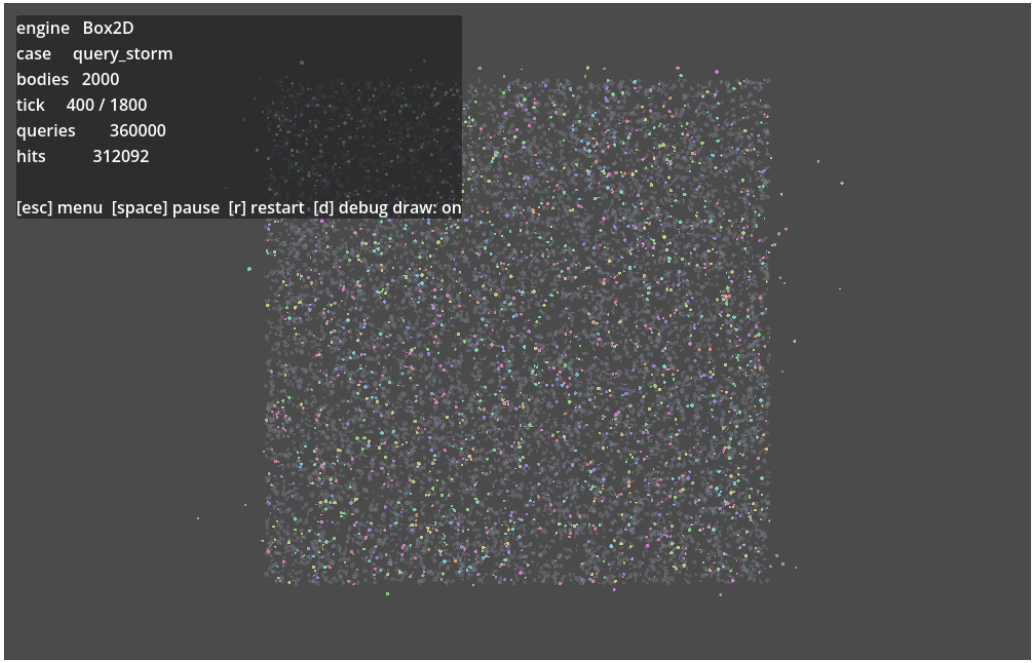
engine	spread
Box2D	7661
Rapier2D	2830

`query_storm` (2d)

Rapier2D



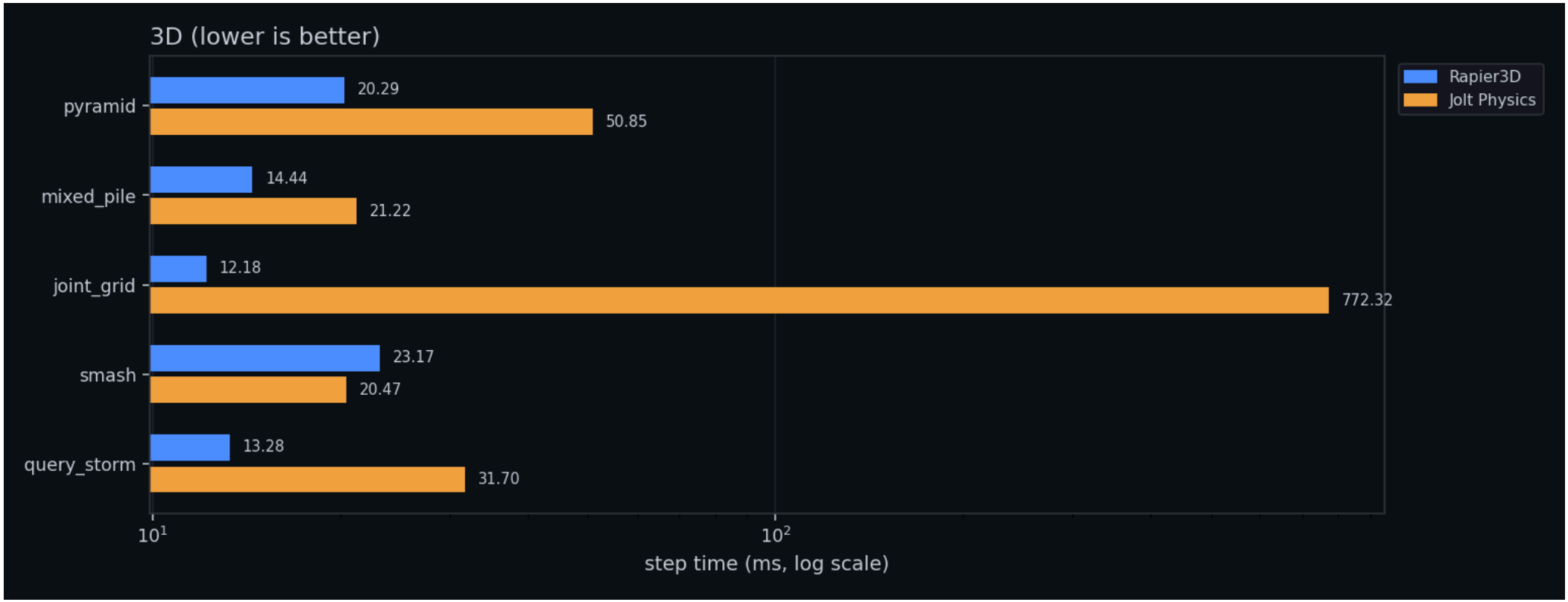
Box2D

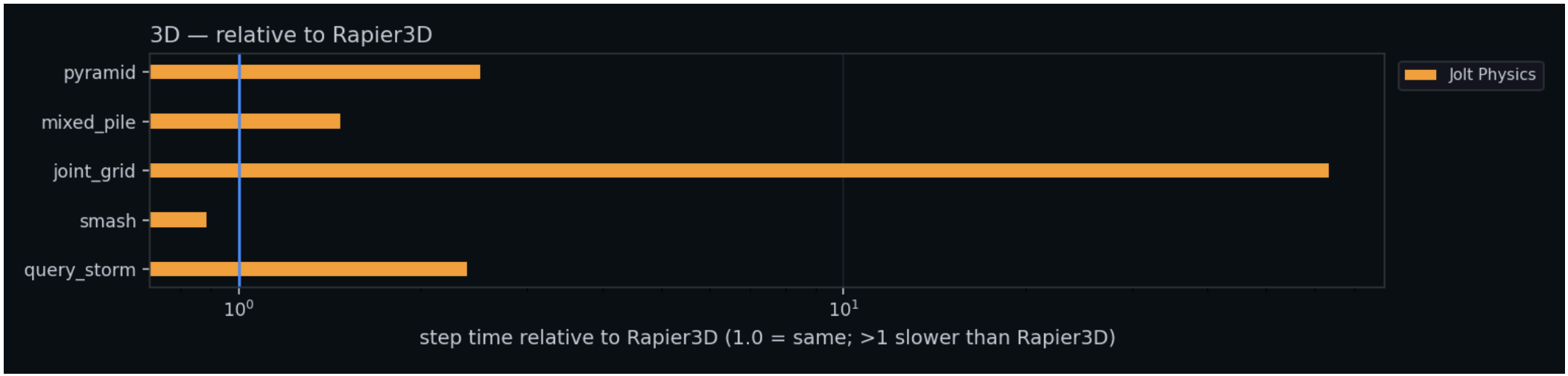


engine	step p50	step p99	vs Rapier2D
Box2D	**5.78 ms**	12.24 ms	0.15x
Rapier2D	39.65 ms	119.01 ms	—

engine	`hits`	`queries`
Box2D	178837	206100
Rapier2D	142100	206100

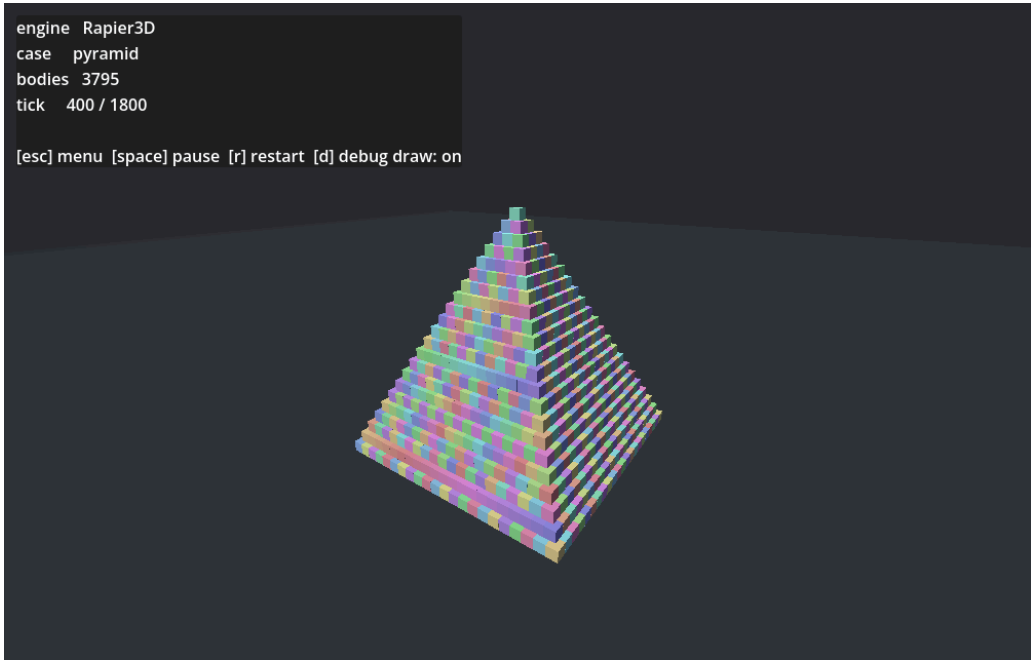
3D (lower is better)



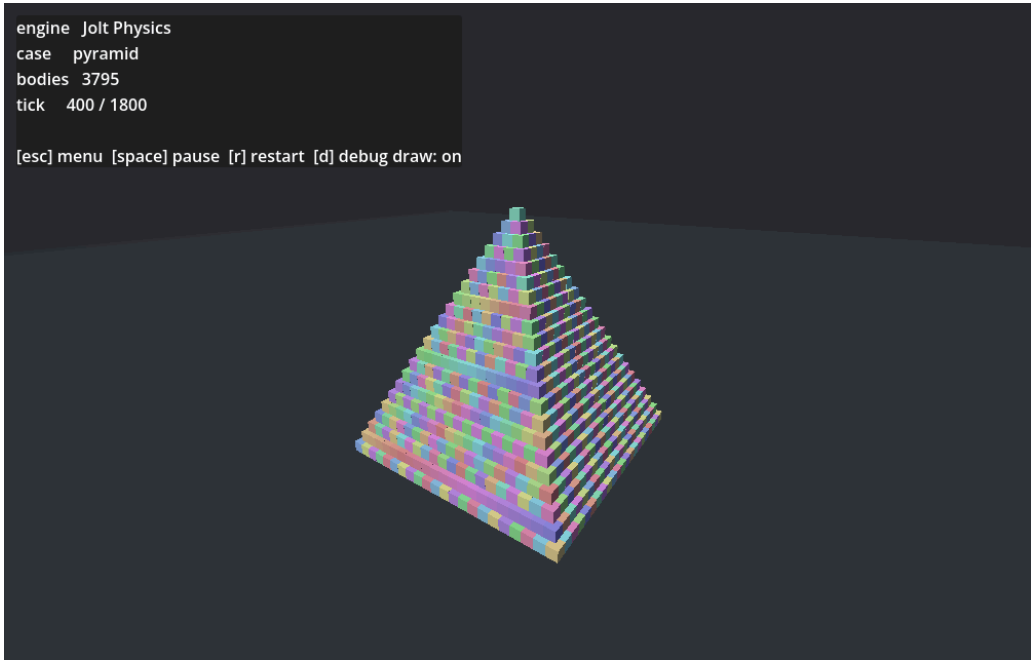


`pyramid` (3d)

Rapier3D



Jolt Physics

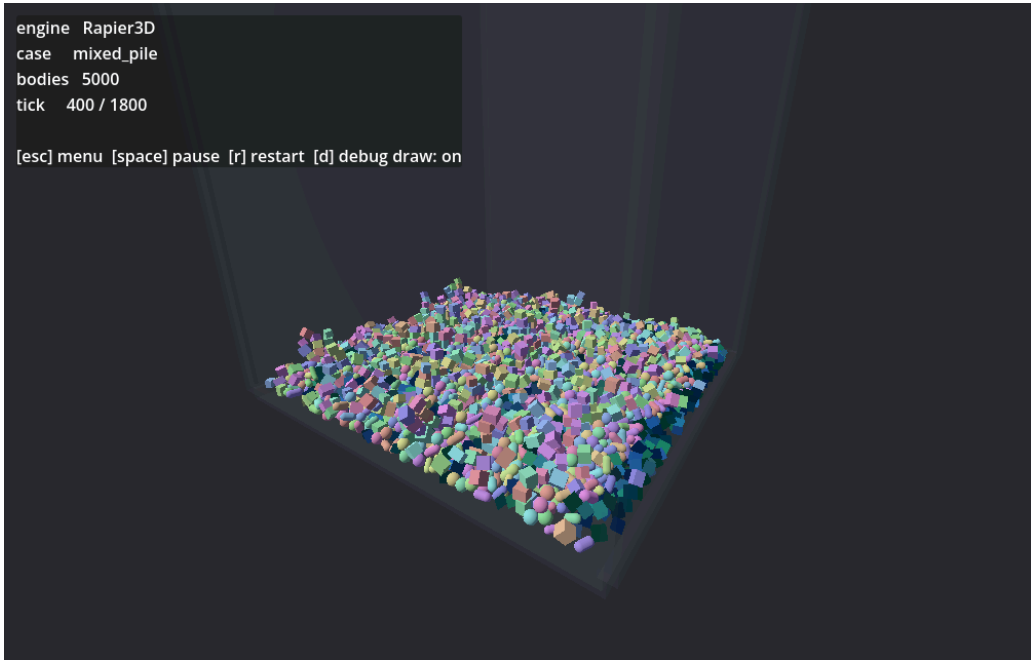


engine	step p50	step p99	vs Rapier3D
Rapier3D	**20.29 ms**	34.32 ms	—
Jolt Physics	50.85 ms	172.37 ms	2.51x

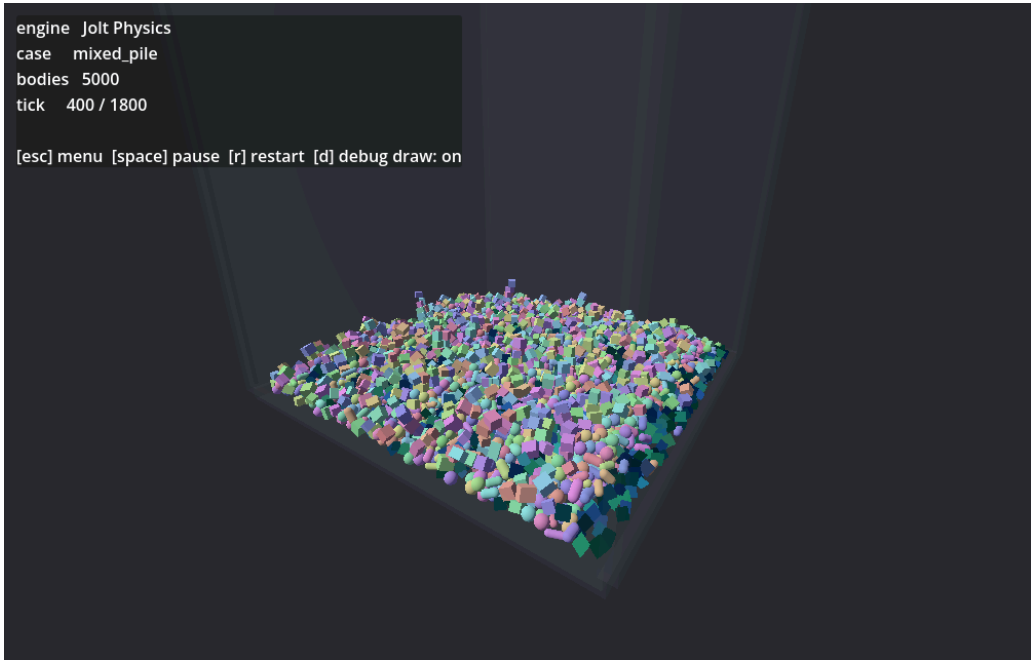
engine	`sync_lag_angle_max_deg`
Jolt Physics	0.05595
Rapier3D	0.05595

`mixed_pile` (3d)

Rapier3D



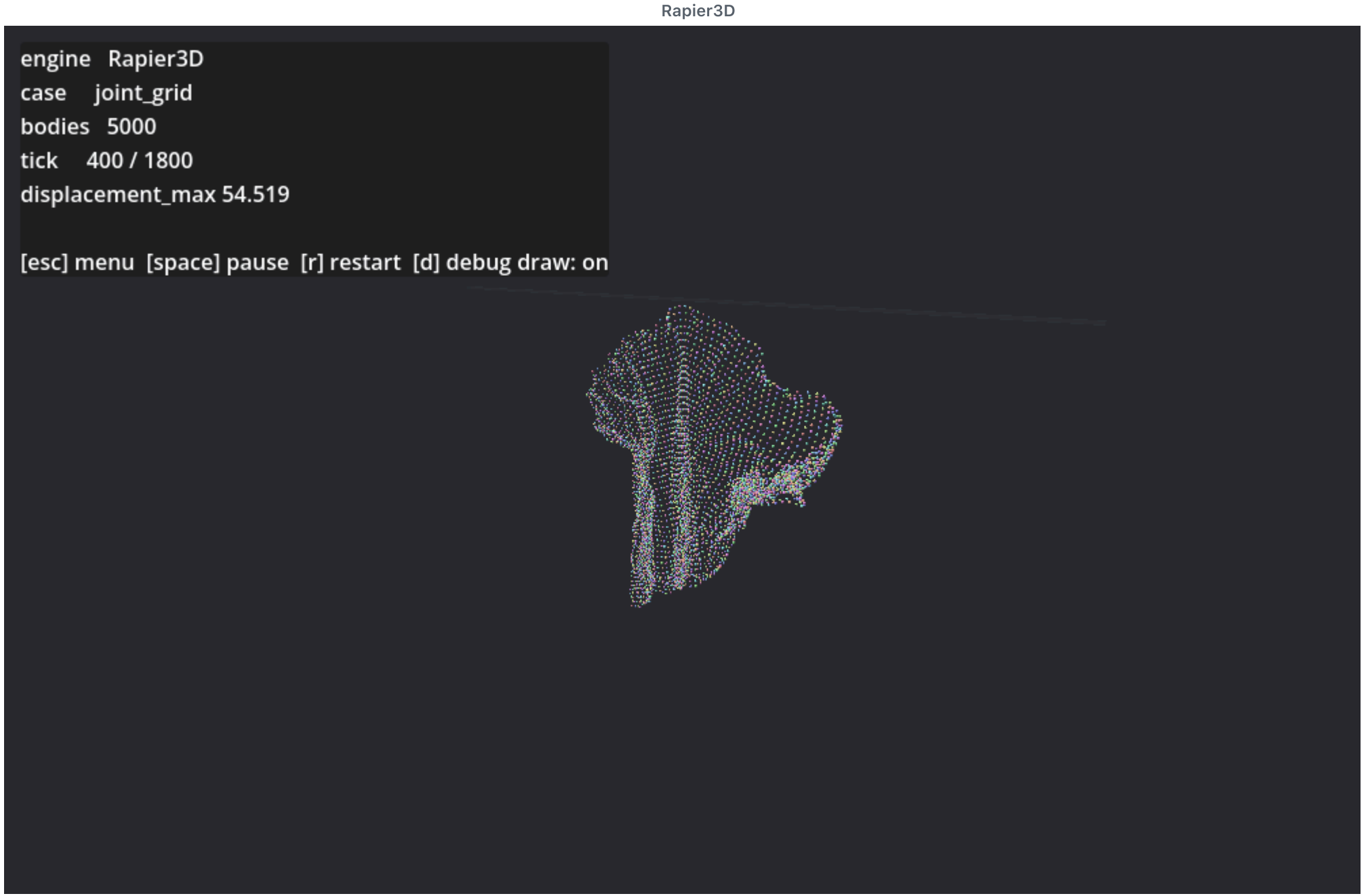
Jolt Physics



engine	step p50	step p99	vs Rapier3D
Rapier3D	**14.44 ms**	23.49 ms	—
Jolt Physics	21.22 ms	37.11 ms	1.47x

engine	`sync_lag_angle_max_deg`
Jolt Physics	0.07913
Rapier3D	0.07913

`joint_grid` (3d)

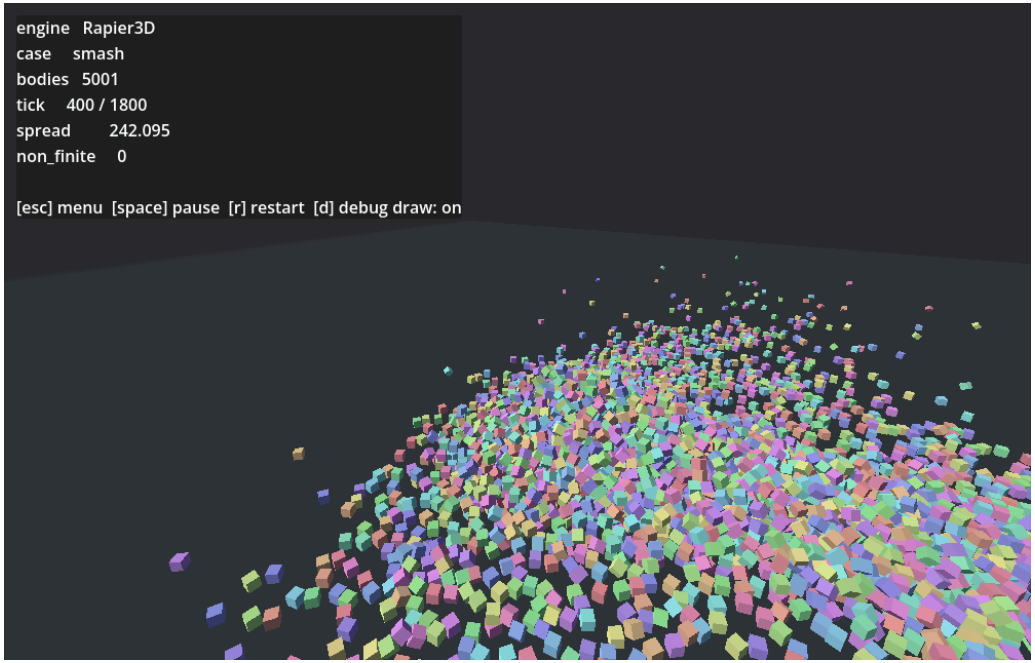


engine	step p50	step p99	vs Rapier3D
Rapier3D	**12.18 ms**	30.75 ms	—
Jolt Physics	772.32 ms	1249.12 ms	63.39x

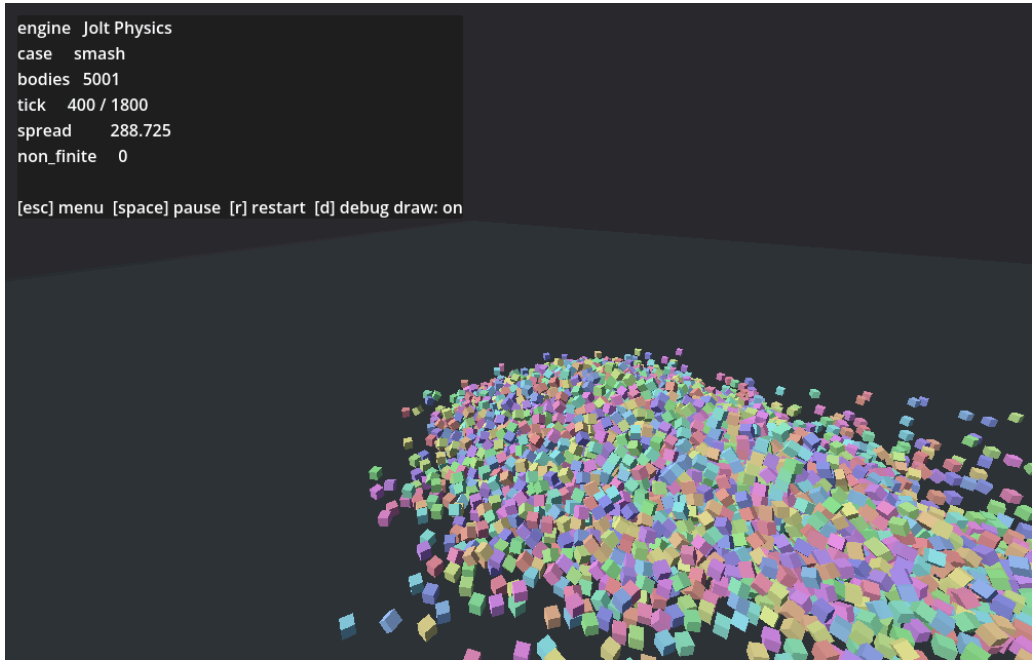
engine	displacement_max	sync_lag_angle_max_deg
Jolt Physics	—	0.06853
Rapier3D	14.92	0.05595

`smash` (3d)

Rapier3D



Jolt Physics

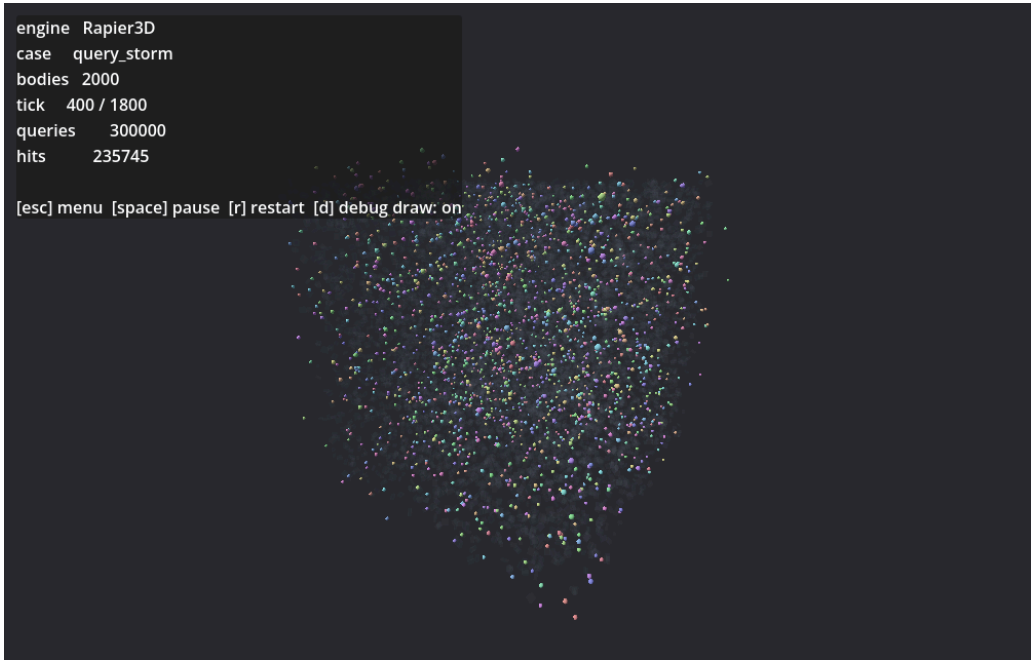


engine	step p50	step p99	vs Rapier3D
Jolt Physics	**20.47 ms**	53.13 ms	0.88x
Rapier3D	23.17 ms	119.45 ms	—

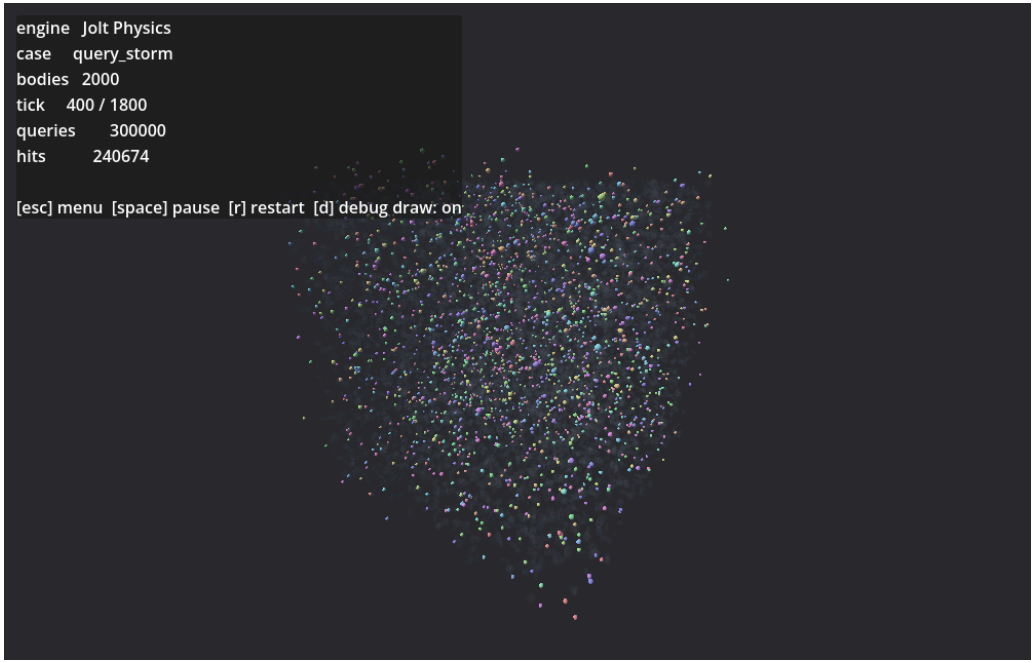
engine	`spread`	`sync_lag_angle_max_deg`
Jolt Physics	148.5	0.06853
Rapier3D	167.4	0.06853

`query_storm` (3d)

Rapier3D



Jolt Physics



engine	step p50	step p99	vs Rapier3D
Rapier3D	**13.28 ms**	16.50 ms	—
Jolt Physics	31.70 ms	69.19 ms	2.39x

engine	hits	queries	sync_lag_angle_max_deg
Jolt Physics	137566	171750	0.05595
Rapier3D	134707	171750	0.05595